

SIH26131

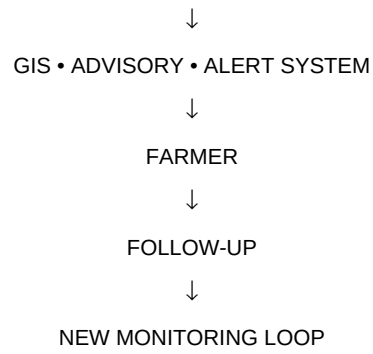
Crop Disease & Pest Detection and Early-Warning Platform

Complete System Architecture • Version 1.0

Core principle: Sensors continuously monitor farm conditions → the Risk Engine detects elevated/rising risk → the relevant camera is triggered → AI analyzes the captured image → low-confidence results go to an expert → final assessment → advisory/alert → follow-up monitoring.

1. Master Architecture





Architectural principle: the camera is event-driven, not continuously AI-analyzed. Sensors provide the first layer of evidence and the Risk Engine decides when visual inspection is required.

2. Farm and Zone Layer

The farm is divided into monitoring zones. Each zone has representative sensing and camera coverage. This enables targeted inspection instead of activating cameras across the whole farm.

Farm	Zone A	Zone B	Zone C	Zone D
Crop / stage / history	Sensors + camera	Sensors + camera	Sensors + camera	Sensors + camera

3. Sensors and Their Benefits

Sensor / device	Measures	Benefit
Air temperature	Temperature and trend	Environmental context for crop-health and disease/pest risk.
Relative humidity	Humidity and duration	Important evidence for humid conditions; duration/trend are more useful than a single reading.
Rain gauge	Rain occurrence / amount	Recent wet-condition context; helps interpret humidity and leaf wetness.
Leaf-wetness sensor	Leaf wetness / duration	Direct crop-surface evidence; useful for disease-risk assessment.
Soil-moisture sensor	Root-zone moisture	Detects abnormal moisture and adds crop/environment context.
Soil-temperature sensor (optional)	Soil temperature	Additional root-zone context; not essential to the first prototype.
Smart pest trap / monitor	Pest type / count / time	Detects rising pest activity and supports early warning.

4. Digital Farm Simulator

For the SIH prototype, physical sensors are not purchased. A software simulator generates realistic time-series telemetry. Simulated readings are explicitly marked as simulated. The same logical telemetry contract is intended for future real devices.

Future real device	SIH prototype
Temperature sensor	Temperature simulator
Humidity sensor	Humidity simulator
Rain gauge	Rain simulator
Soil sensor	Soil simulator
Leaf-wetness sensor	Leaf-wetness simulator
Pest trap	Pest simulator
Physical camera	Camera trigger + test/provided image

5. Data Validation

Raw readings are checked before intelligence uses them. Validation covers impossible ranges, missing readings, stale data, device-offline status, timestamp errors and duplicates. Data provenance is retained.

6. Feature Engine

The Feature Engine converts raw readings into meaningful temporal/environmental indicators so the Risk Engine is not dependent on a single sensor reading.

Raw input	Derived features
Humidity	Average, trend, duration above configured level, rate of change
Rainfall	Recent accumulation, rain event, time since rain
Leaf wetness	Wetness duration, repeated wetness
Soil moisture	Trend and abnormal-moisture condition
Pest count	Moving average, growth rate, spike and trend

7. Risk Engine

The Risk Engine is the decision-making point before visual inspection. It combines environmental evidence, pest activity, crop context, historical observations and risk trends.

Inputs	Outputs
Temperature • Humidity • Rainfall • Leaf wetness • Soil moisture • Pest activity • Crop • Crop stage • History • Trends	Risk score • Risk level • Risk trend • Explainable reasons

8. Camera Trigger and Targeting

When a zone reaches a calibrated inspection condition, the system identifies that zone and triggers its camera. The threshold should be selected using validation rather than an arbitrary fixed number.

Example: Zone A = low, Zone B = low, Zone C = high/rising → only Zone C camera is triggered.

9. Camera and Image Processing

The intended real-world visual device is a fixed RGB field camera positioned to observe a defined crop zone. It is event-triggered by the Risk Engine.

Step	Operation
1	Risk Engine identifies risky zone
2	Camera for that zone is triggered
3	Image is captured
4	Image quality is checked
5	Poor image → recapture; good image → AI Vision

Camera benefits

- Provides direct visual evidence of crop symptoms.
- Confirms whether a sensor-driven risk signal corresponds to visible crop conditions.
- Supports disease/pest identification and severity estimation.
- Avoids unnecessary continuous image collection and AI inference.
- Enables targeted inspection.

10. AI Vision Engine

The AI analyzes the triggered image for disease, pest and severity. Predictions retain confidence and model-version metadata.

AI task	Output
Disease detection	Disease class / probability
Pest detection	Pest class / probability
Severity	Severity class or supported affected-area estimate
Model metadata	Confidence and model version

11. Confidence Gate and Expert Workflow

AI confidence determines whether the result can continue automatically or requires human verification. The threshold must be calibrated from validation data.

HIGH CONFIDENCE → continue to assessment

LOW CONFIDENCE → EXPERT REVIEW → Confirm / Correct / Unable to Determine → Final Assessment

The expert receives the image, AI prediction, confidence, severity, risk score, sensor conditions and relevant history. The farmer is not required to intervene during normal automated monitoring.

12. GIS / Zone Intelligence

GIS links observations and risk to farm zones. Its purpose is targeted monitoring and spatial understanding; there is no officer-management workflow.

Zone	Risk	Purpose
Zone A	Low	Continue monitoring
Zone B	Moderate	Watch trend
Zone C	High	Camera inspection / hotspot
Zone D	Low	Continue monitoring

13. Advisory Engine

After a sufficiently reliable assessment, the Advisory Engine uses crop, disease/pest condition, severity, risk and crop stage to retrieve validated guidance. It should use a controlled agricultural knowledge base rather than invent treatment instructions.

14. Alert System

The farmer receives alerts for high risk, detected conditions, required expert verification and significant changes during follow-up.

Event	Example
High risk	High crop-health risk detected in Zone C. Camera inspection initiated.
Confirmed condition	Disease/pest condition detected in Zone C with severity information.
Expert required	AI could not confidently identify the crop condition. Expert verification required.

15. Follow-up and Closed Loop

Monitoring continues after an alert or assessment. New sensor data and, when needed, another camera inspection are compared with previous observations to determine whether risk is improving or worsening.

Detection → Alert → Continued monitoring → Risk recalculation → Camera if necessary → New assessment → Comparison → History

16. Database / Data Model

Entity	Purpose
Users	Farmer and expert accounts
Farms	Farm identity and configuration
Zones	Spatial monitoring units
Crops	Crop and crop-stage information
Devices	Sensor and camera registration
Sensor readings	Time-series telemetry and provenance
Camera events	Trigger and capture records
Images	Captured/associated crop images
AI predictions	Disease/pest predictions, confidence and model version
Risk events	Risk score, level, trend and reasons
Expert reviews	Confirmation/correction outcomes
Alerts	Notifications and event status
Advisories	Validated guidance delivered to farmer
Follow-ups	Subsequent observations and outcomes

17. Real Farm vs SIH Prototype

Real deployment	SIH prototype
Physical sensors	Digital sensor simulator
IoT communication	Simulator → telemetry API
Physical camera	Simulated trigger + prepared/test image
Real crop image	Dataset / captured demo image
Same Risk Engine	Same Risk Engine
Same AI pipeline	Same AI pipeline
Same expert workflow	Same expert workflow
Same farmer application	Same farmer application

18. Final Architecture Philosophy

Component	Question answered
Sensors	What are the current field conditions?
Feature Engine	What patterns and trends exist?
Risk Engine	Does this zone need visual inspection?
Camera	What is visually happening in the risky zone?
AI Vision	What disease/pest might be present and how severe is it?
Confidence Gate	Is AI confident enough?
Expert	Can the uncertain result be confirmed/corrected?

Component	Question answered
GIS	Where is the risk occurring?
Advisory	What validated guidance applies?
Alerts	Who needs to know?
Follow-up	Is the situation improving or worsening?

Final intelligence loop: SENSE → VALIDATE → BUILD FEATURES → ASSESS RISK → TARGET CAMERA → ANALYZE IMAGE → CONFIDENCE CHECK → EXPERT IF NEEDED → FINAL ASSESSMENT → ADVISORY + ALERT → FOLLOW-UP → NEW MONITORING.